🔧 Docker Compose

▯ Overview & Use Cases

What is Docker Compose?
Docker Compose is a tool that allows you to **define and run multi-container applications** using a YAML file. Instead of writing long docker run commands for each container, you define everything once in a file and start your entire app stack with one command.

Why use it?

- Simplifies setup of apps that need multiple containers (e.g., web + database).
- Useful for development, testing, and CI/CD.
- Makes environments reproducible and shareable.

Real-world examples:

- Web app + MySQL + Redis
- Backend API + MongoDB + Nginx reverse proxy

▯ Simplifying Multi-Container Setup

Without Compose:

```
docker network create my-net



docker run -d --name db --network my-net mysql

docker run -d --name web --network my-net -p 8080:80 nginx
```

This is error-prone and hard to maintain.

With Docker Compose:

```
Docker compose up -d
```

A single command starts the whole app with correct networking, environment variables, volumes, etc.

⚙ Demo Using Docker Compose

Create a folder, e.g., myapp/, and inside it create this file:

docker-compose.yml

```
version: '3'

services:
  db:
    image: mysql:5.7
    environment:
      MYSQL_ROOT_PASSWORD: root
    volumes:
      - db_data:/var/lib/mysql

  web:
    image: nginx
    ports:
      - "8080:80"
    depends_on:
      - db

volumes:
  db_data:
```

Run the application:

```
docker-compose up -d
```

Check services:

```
docker-compose ps
```

Access the app:
Visit http://localhost:8080 in your browser (Nginx page will appear).

Stop and remove services:

```
docker-compose down
```

🛠 Writing a docker-compose.yml for Multi-Service Applications

Key sections in docker-compose.yml:

Section	Description
services:	Defines each container (web, db, etc.)
image:	Docker image to use
build:	Path to build custom Dockerfile
ports:	Maps host and container ports

volumes:	For persistent storage
depends_on:	Ensures service start order
networks:	(Optional) define custom networks

📋 Docker Compose Commands

Command	Description
docker-compose up	Start all services
docker-compose up -d	Start in detached mode
docker-compose down	Stop and remove all containers and networks
docker-compose logs	View logs from all services
docker-compose ps	Check status of running containers
docker-compose build	Build services defined with Dockerfile
docker-compose up --scale web=3	Run 3 replicas of the web service

---

🔁 Scaling Services (Basic)

To scale a service (like web) to multiple instances:

docker-compose up --scale web=3 -d

Works with Compose version 3+ when running in **Swarm mode** (or use version 2 for local scaling).

✅ Summary

- Docker Compose helps run multi-container setups easily.
- Use a YAML file to define your application stack.
- Everything is version-controlled, portable, and reproducible.
- One command can start, stop, or scale your entire app.